

Phần 1: Các lệnh cơ bản thao tác với Docker

```
C:\Users\nguye>docker --version
Docker version 29.2.1, build a5c7197
```

```
C:\Users\nguye>docker run hello-world
```

Hello from Docker!

This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:

1. The Docker client contacted the Docker daemon.
2. The Docker daemon pulled the "hello-world" image from the Docker Hub. (amd64)
3. The Docker daemon created a new container from that image which runs the executable that produces the output you are currently reading.
4. The Docker daemon streamed that output to the Docker client, which sent it to your terminal.

To try something more ambitious, you can run an Ubuntu container with:

```
$ docker run -it ubuntu bash
```

Share images, automate workflows, and more with a free Docker ID:

<https://hub.docker.com/>

For more examples and ideas, visit:

<https://docs.docker.com/get-started/>

```
C:\Users\nguye>docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
```

```
9eef040df109: Pull complete
```

```
206356c42440: Pull complete
```

```
df9da45c1db2: Pull complete
```

```
a9d395129dce: Pull complete
```

```
18a071c04bd1: Pull complete
```

```
79697674b897: Pull complete
```

```
75ald70aee50: Pull complete
```

```
d99947bc9177: Download complete
```

```
23abb0f9ce55: Download complete
```

```
Digest: sha256:bc45d248c4e1d1709321de61566eb2b64d4f0e32765239d66573666be7f13349
```

```
Status: Downloaded newer image for nginx:latest
```

```
docker.io/library/nginx:latest
```

```
C:\Users\nguye>docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
hello-world:latest	85404b3c5395	25.9kB	9.52kB	
nginx:latest	bc45d248c4e1	240MB	65.8MB	U

Info → U In Use

```
C:\Users\nguye>docker run -d nginx
```

```
9996cbd05ce3f0798acb01cf60c5ce6e56ce5eca926a59762144df289c12cf2f
```

```
C:\Users\nguye>docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS        NAMES
9996cbd05ce3   nginx    "/docker-entrypoint...." 19 seconds ago Up 18 seconds 80/tcp       awesome_wright
```

```
C:\Users\nguye>docker ps -a
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS        NAMES
9996cbd05ce3   nginx    "/docker-entrypoint...." 48 seconds ago Up 48 seconds 80/tcp       awesome_wright
197878d433df   hello-world "/hello"                3 minutes ago Exited (0) 3 minutes ago
lucid_ganguly
a20ee9c670f9   hello-world "/hello"                3 minutes ago Exited (0) 3 minutes ago
lucid_gauss
```

```
C:\Users\nguye>docker logs 9996cbd05ce3
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/03/12 14:57:44 [notice] 1#1: using the "epoll" event method
2026/03/12 14:57:44 [notice] 1#1: nginx/1.29.6
2026/03/12 14:57:44 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/03/12 14:57:44 [notice] 1#1: OS: Linux 6.6.87.2-microsoft-standard-WSL2
2026/03/12 14:57:44 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2026/03/12 14:57:44 [notice] 1#1: start worker processes
2026/03/12 14:57:44 [notice] 1#1: start worker process 29
2026/03/12 14:57:44 [notice] 1#1: start worker process 30
2026/03/12 14:57:44 [notice] 1#1: start worker process 31
2026/03/12 14:57:44 [notice] 1#1: start worker process 32
2026/03/12 14:57:44 [notice] 1#1: start worker process 33
2026/03/12 14:57:44 [notice] 1#1: start worker process 34
2026/03/12 14:57:44 [notice] 1#1: start worker process 35
2026/03/12 14:57:44 [notice] 1#1: start worker process 36
2026/03/12 14:57:44 [notice] 1#1: start worker process 37
2026/03/12 14:57:44 [notice] 1#1: start worker process 38
2026/03/12 14:57:44 [notice] 1#1: start worker process 39
2026/03/12 14:57:44 [notice] 1#1: start worker process 40
```

```
C:\Users\nguye>docker exec -it 9996cbd05ce3 /bin/sh
# |
```

```
# ls
bin  dev          docker-entrypoint.sh  home  lib64  mnt  proc  run  srv  tmp  var
boot docker-entrypoint.d  etc          lib   media  opt  root  sbin  sys  usr
# |
```

```
C:\Users\nguye>docker stop 9996cbd05ce3
9996cbd05ce3
```

```
C:\Users\nguye>docker restart 9996cbd05ce3
9996cbd05ce3
```

```
C:\Users\nguye>docker stop 9996cbd05ce3
9996cbd05ce3
```

```
C:\Users\nguye>docker rm 9996cbd05ce3
9996cbd05ce3
```

```
C:\Users\nguye>docker container prune
WARNING! This will remove all stopped containers.
Are you sure you want to continue? [y/N] y
Deleted Containers:
197878d433df0652487ec43c11ed41a9a9dfdf0e54897e318d22067be3c00d7e
a20ee9c670f9f884b30d5581bf6666a41bfcec5c9b1e65a309c1dbab01cbf5b6

Total reclaimed space: 8.192kB
```

```
C:\Users\nguye>docker rmi bc45d248c4e1
Untagged: nginx:latest
Deleted: sha256:bc45d248c4e1d1709321de61566eb2b64d4f0e32765239d66573666be7f13349
```

```
C:\Users\nguye>docker image prune -a
WARNING! This will remove all images without at least one container associated to them.
Are you sure you want to continue? [y/N] y
Deleted Images:
untagged: hello-world:latest
deleted: sha256:85404b3c53951c3ff5d40de0972b1bb21faf2a2e8daa235355baf44f33db9dbdd
deleted: sha256:2771e37a12b7bcb2902456ecf3f29bf9ee11ec348e66e8eb322d9780ad7fc2df
deleted: sha256:6b75187531c5e9b6a85c8946d5d82e4ef3801e051fbff338f382f3edfa60e3d2

Total reclaimed space: 21.86kB
```

```
C:\Users\nguye>docker run -d -p 8080:80 nginx
a2d23e05ba645fbac972d404c851e17b1a58ecd298c1ed768e130c147875a0bd
```

```
C:\Users\nguye>docker inspect a2d23e05ba64
[
  {
    "Id": "a2d23e05ba645fbac972d404c851e17b1a58ecd298c1ed768e130c147875a0bd",
    "Created": "2026-03-12T15:11:55.236075378Z",
    "Path": "/docker-entrypoint.sh",
    "Args": [
      "nginx"
    ]
  }
]
```

```
C:\Users\nguye>docker run -d -v mydata:/data nginx
e4861bb4785f10adfa15aceac8d33e40f146079cf2142e165e56dfdf487c24f6
```

```
C:\Users\nguye>docker volume ls
DRIVER      VOLUME NAME
local      mydata
```

```
C:\Users\nguye>docker volume prune
WARNING! This will remove anonymous local volumes not used by at least one container.
Are you sure you want to continue? [y/N] y
Total reclaimed space: 0B
```

```
C:\Users\nguye>docker run -d --name my_nginx nginx
974e9ca254becbbf6974b53aac440005c664e7c4a49408000cfb23ec0de4d187
```

```
C:\Users\nguye>docker network ls
NETWORK ID          NAME                DRIVER              SCOPE
2d0f4d1536cf        bridge             bridge             local
fd8966bd7bdc        host               host               local
8acf1d2a1c54        none               null               local
```

```
C:\Users\nguye>docker network create my_network
d1d3a5080025b78d5a79166a6f45a66a6e222c811c8421dc7b45b5ecb1a71d27
```

```
C:\Users\nguye>docker run -d --network my_network --name my_container nginx
ca2e489e4979e9f7aee78e2c703877ef463e66336e9e3b825564e43147553ea7
```

```
C:\Users\nguye>docker run -d -e MY_ENV=hello_world nginx
a32b88d90c932ea2e188d88e893c1cf9a2c3d436e2a059b280305298d4839c1c
```

```
C:\Users\nguye>docker logs -f my_nginx
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/03/12 15:14:21 [notice] 1#1: using the "epoll" event method
2026/03/12 15:14:21 [notice] 1#1: nginx/1.29.6
2026/03/12 15:14:21 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/03/12 15:14:21 [notice] 1#1: OS: Linux 6.6.87.2-microsoft-standard-WSL2
2026/03/12 15:14:21 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2026/03/12 15:14:21 [notice] 1#1: start worker processes
2026/03/12 15:14:21 [notice] 1#1: start worker process 29
2026/03/12 15:14:21 [notice] 1#1: start worker process 30
2026/03/12 15:14:21 [notice] 1#1: start worker process 31
2026/03/12 15:14:21 [notice] 1#1: start worker process 32
2026/03/12 15:14:21 [notice] 1#1: start worker process 33
2026/03/12 15:14:21 [notice] 1#1: start worker process 34
2026/03/12 15:14:21 [notice] 1#1: start worker process 35
2026/03/12 15:14:21 [notice] 1#1: start worker process 36
2026/03/12 15:14:21 [notice] 1#1: start worker process 37
2026/03/12 15:14:21 [notice] 1#1: start worker process 38
2026/03/12 15:14:21 [notice] 1#1: start worker process 39
2026/03/12 15:14:21 [notice] 1#1: start worker process 40
```

```
C:\Users\nguye>cd Desktop

C:\Users\nguye\Desktop>mkdir docker-nginx
A subdirectory or file docker-nginx already exists

C:\Users\nguye\Desktop>cd docker-nginx
```

```
C:\Users\nguye\Desktop\docker-nginx>dir
Volume in drive C is OS
Volume Serial Number is AED3-3B81

Directory of C:\Users\nguye\Desktop\docker-nginx

03/12/2026  10:22 PM    <DIR>          .
03/12/2026  10:19 PM    <DIR>          ..
03/12/2026  10:22 PM    <DIR>          docker-nginx
03/12/2026  10:22 PM                60 Dockerfile
03/12/2026  10:20 PM                54 index.html
                2 File(s)              114 bytes
                3 Dir(s)  46,932,946,944 bytes free
```

```
C:\Users\nguye\Desktop\docker-nginx>docker build -t my_nginx_image .
[+] Building 0.7s (7/7) FINISHED
=> [internal] load build definition from Dockerfile                                0.1s
=> => transferring dockerfile: 97B                                                0.0s
=> [internal] load metadata for docker.io/library/nginx:latest                  0.1s
=> [internal] load .dockerignore                                                  0.0s
=> => transferring context: 2B                                                    0.0s
=> [internal] load build context                                                 0.0s
=> => transferring context: 91B                                                  0.0s
=> [1/2] FROM docker.io/library/nginx:latest@sha256:bc45d248c4e1d1709321de61566eb2b64d4f0e32765239d66573666be7f13349 0.1s
=> => resolve docker.io/library/nginx:latest@sha256:bc45d248c4e1d1709321de61566eb2b64d4f0e32765239d66573666be7f13349 0.0s
=> [2/2] COPY index.html /usr/share/nginx/html/index.html                      0.0s
=> exporting to image                                                            0.2s
=> => exporting layers                                                            0.1s
=> => exporting manifest sha256:419fbb782104ad9f111d890473a8b0b5f86299eced559b352eb8bd6be035f55e          0.0s
=> => exporting config sha256:014d516353fcfa7e50c8608f16a5604de6bfb2705c1335bb81f064cf1ad2c202          0.0s
=> => exporting attestation manifest sha256:77efbe55fcdabfbde8edc78155fd00f93071543f28eed8e9587d4a6b08960a75    0.0s
=> => exporting manifest list sha256:0af70d946bbd0c09b2064b42f47f03fcfd837d0c04cd0a3818d3d537d292b209        0.0s
=> => naming to docker.io/library/my_nginx_image:latest                          0.0s
=> => unpacking to docker.io/library/my_nginx_image:latest                       0.0s
```

```
C:\Users\nguye\Desktop\docker-nginx>docker run -d -p 8080:80 my_nginx_image
4479f8e9fc03b20a295e78aec3f716eec9d467c29959ef01af8ba624d6b7d66e
```